

HomeSOClab — Network Architecture (Phase 1, Layer 2)

This document describes the segmented network design for the HomeSOClab. The design follows a **zero-trust-inspired segmentation model**: every zone is its own VLAN/L2 domain, and the firewall is the single inter-VLAN router enforcing an explicit allow-list policy (default deny).

1. Design Goals & Security Rationale

Goal	How it is achieved
Contain malware/exploits to the lab	Lab VLANs cannot reach the home LAN (192.168.1.0/24)
Realistic attack path	Attacker VLAN → Victim VLAN allowed; Victim cannot initiate back
Prevent victim exfiltration / botnet C2	Victim VLAN has no internet (default-deny WAN)
Full packet visibility for detection	Port mirroring (SPAN) copies inter-VLAN traffic to the SIEM sensor
Safe administration	Management VLAN is the only plane allowed to reach Proxmox/pfSense/SIEM admin UIs
Blast-radius reduction	Default-deny between every pair of zones; only documented flows are permitted

2. VLAN / IP Addressing Scheme

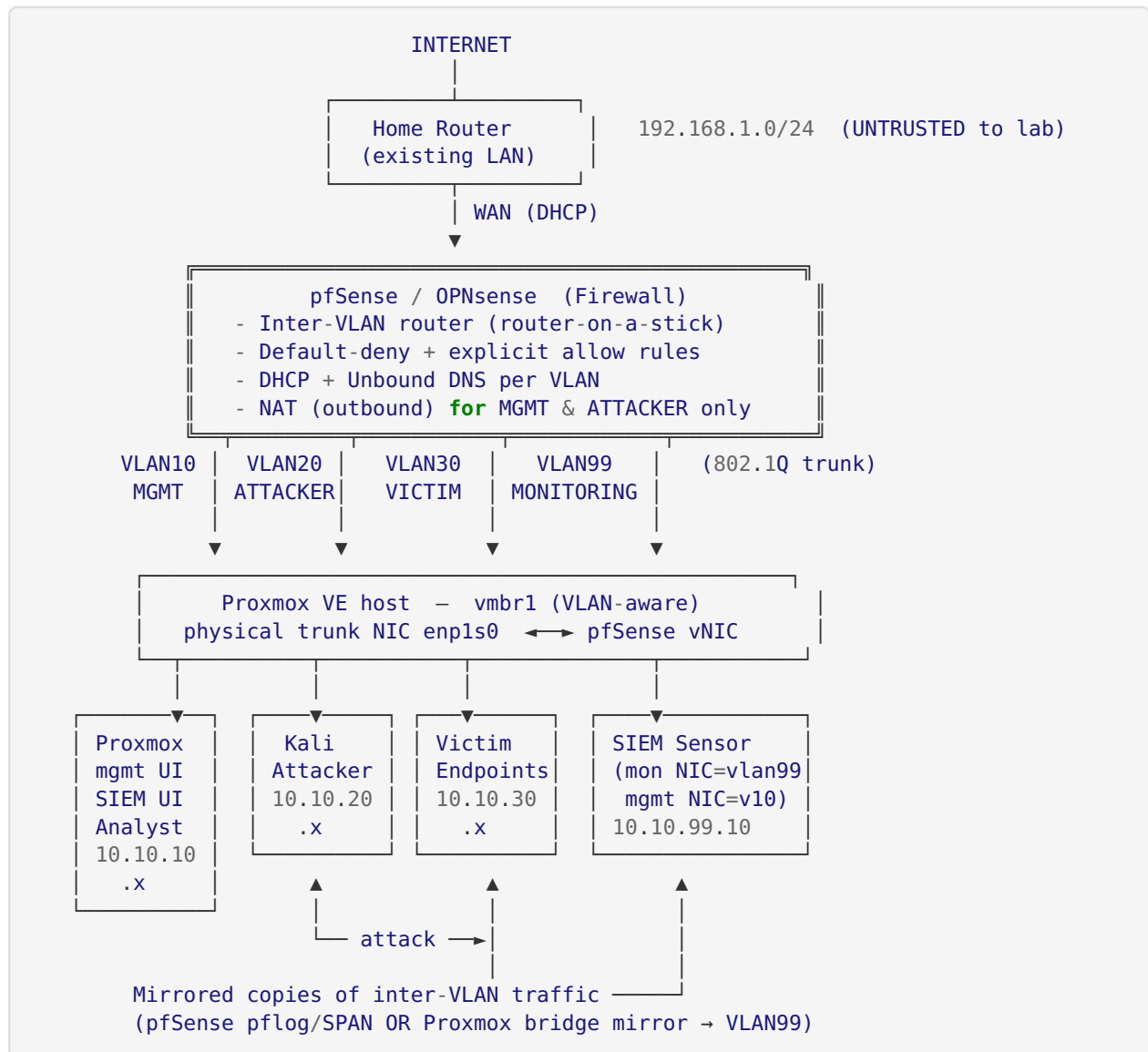
VLAN ID	Zone	Subnet (CIDR)	Gateway (pfSense)	DHCP Pool	Internet	Purpose
10	Management	10.10.10.0 /24	10.10.10.1	.100-.199	Controlled	Proxmox, pfSense, SIEM UI, analyst WS
20	Attacker	10.10.20.0 /24	10.10.20.1	.100-.199	✓ Yes	Kali / red-team tooling
30	Victim	10.10.30.0 /24	10.10.30.1	.100-.199	✗ No	Vulnerable endpoints / targets
99	Monitoring	10.10.99.0 /24	10.10.99.1	static only	✗ No	SPAN sink + SIEM sensor sniff NIC

Reserved static addresses (outside DHCP pools)

Host	IP	VLAN
pfSense/OPNsense (router)	10.10.X.1 per VLAN	all
Proxmox pve01 mgmt	10.10.10.2	10
SIEM management UI	10.10.10.10	10
Analyst workstation	10.10.10.20	10
SIEM sensor (mirror sink NIC)	10.10.99.10	99

Why a separate Monitoring VLAN (99)? The SIEM sensor's sniffing interface receives mirrored traffic and must never source traffic onto production zones. Its management interface lives on VLAN 10 so analysts can reach the dashboard. This dual-NIC pattern mirrors how Security Onion / Wazuh are deployed in production (a management NIC + a promiscuous monitor NIC).

3. Topology Diagram



4. Traffic Flow Matrix (Firewall Policy Summary)

✓ = allowed, ✗ = blocked (default deny), ⚠ = restricted to specific ports

From ↓ \ To →	MGMT	ATTACK- ER	VICTIM	MONIT- ORING	Home LAN	Internet
MGMT	—	✓	✓	⚠	✗	✓
ATTACK- ER	✗	—	✓	✗	✗	✓
VICTIM	✗	✗	—	✗	✗	✗
MONIT- ORING	✗	✗	✗	—	✗	✗

Key policy decisions:

- **Management → everything**: admins/SIEM must reach all zones for triage.
- **Attacker → Victim only**: realistic kill-chain; attacker cannot touch the admin plane or the home LAN (prevents the lab from becoming a pivot point).
- **Victim → nothing outbound**: blocks C2/exfil; only responds to attacker.
- **Monitoring isolated**: sensor only receives mirrored frames; it never routes. Its mgmt UI is reached from VLAN10, not from VLAN99.

See `pfsense/rules/` for the concrete rule definitions per interface.

5. Traffic Mirroring (Network Telemetry Feed)

Phase 1 supports **two complementary mirroring approaches** (see `pfsense/port-mirror.md` and `proxmox/bridge-mirror.md`):

1. **Proxmox bridge mirroring (recommended, hypervisor-level)**: a `tc`-based SPAN that copies all frames traversing `vmbr1` to a dedicated mirror port connected to the SIEM sensor's monitor NIC (VLAN 99). Captures all VM traffic including intra-VLAN east-west traffic.
2. **pfSense span/packet capture (firewall-level)**: mirrors traffic crossing the firewall to a monitor interface. Simpler but only sees inter-VLAN/WAN traffic, not intra-VLAN.

Both deliver line-rate copies to `10.10.99.10` where the SIEM sensor runs in promiscuous mode (Zeek/Suricata) in later phases.